

Topic 14: Mechanical Oscillations, Damping & Pure Resonance

Professor Leonard Master Series

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: The Spring-Mass-Damper Universe

Professor Leonard brings 2nd-order ODEs to life with Newtonian mechanics:

"Newton's 2nd Law: $ma = \sum F$. A mass on a spring experiences Restoring Force $-kx$, Damping Friction $-cx'$, and External Forcing $F_0 \cos(\omega t)$. Summing them up gives the master equation of mechanical engineering:"

$$m \frac{d^2 x}{dt^2} + c \frac{dx}{dt} + kx = F_0 \cos(\omega t)$$

Characteristic Equation: $mr^2 + cr + k = 0$. Discriminant $\Delta = c^2 - 4mk$:

- $c^2 - 4mk < 0$: **Underdamped** (Oscillates with exponentially decaying envelope).
- $c^2 - 4mk = 0$: **Critically Damped** ($c_{\text{crit}} = 2\sqrt{km}$, returns to rest fastest without oscillating).
- $c^2 - 4mk > 0$: **Overdamped** (Sluggish, no oscillations).

2. Professor Leonard's Whiteboard Problem Walkthroughs

Problem 14.1: Underdamped Motion & Amplitude-Phase Form

Statement: A 2-kg mass is attached to a spring with $k = 50$ N/m and a dashpot with $c = 12$ N·s/m. Released from $x(0) = 0.2$ m with initial velocity $v(0) = -1$ m/s. Find $x(t)$ in amplitude-phase form $Ae^{-\gamma t} \cos(\omega_d t - \phi)$.

Step-by-Step Whiteboard Solution:

- Step 1: Set up ODE:**

$$2x'' + 12x' + 50x = 0 \implies x'' + 6x' + 25x = 0$$

- Step 2: Characteristic Roots:**

$$r^2 + 6r + 25 = 0 \implies r = \frac{-6 \pm \sqrt{36 - 100}}{2} = -3 \pm 4i$$

Damping factor $\gamma = 3$, Quasi-frequency $\omega_d = 4$ rad/s.

- Step 3: General Solution:**

$$x(t) = e^{-3t}(c_1 \cos 4t + c_2 \sin 4t)$$

- Step 4: Apply Initial Conditions:**

$$x(0) = c_1 = 0.2 = \frac{1}{5}$$

$$x'(t) = -3e^{-3t}(c_1 \cos 4t + c_2 \sin 4t) + e^{-3t}(-4c_1 \sin 4t + 4c_2 \cos 4t)$$

$$x'(0) = -3c_1 + 4c_2 = -1 \implies -3(0.2) + 4c_2 = -1 \implies -0.6 + 4c_2 = -1 \implies 4c_2 = -0.4 \implies c_2 = -0.1$$

$$x(t) = e^{-3t}(0.2 \cos 4t - 0.1 \sin 4t)$$

- **Step 5: Convert to Amplitude-Phase Form:**

$$A = \sqrt{c_1^2 + c_2^2} = \sqrt{(0.2)^2 + (-0.1)^2} = \sqrt{0.04 + 0.01} = \sqrt{0.05} = \frac{\sqrt{5}}{10} \approx 0.2236 \text{ m}$$

$$\phi = \arctan\left(\frac{c_2}{c_1}\right) = \arctan\left(\frac{-0.1}{0.2}\right) = \arctan(-0.5) \approx -0.4636 \text{ rad}$$

$$x(t) = 0.224e^{-3t} \cos(4t + 0.464) \text{ m}$$

Problem 14.2: Pure Resonance Catastrophe (The Bridge Destroyer)

Statement: Solve $x'' + 9x = 4 \cos 3t$ with $x(0) = 0, x'(0) = 0$.

Step-by-Step Whiteboard Solution:

- **Step 1: Natural frequency:** $\omega_0^2 = 9 \implies \omega_0 = 3 \text{ rad/s}$. Driving frequency $\omega = 3 \text{ rad/s}$. $\omega = \omega_0 \implies$ **PURE RESONANCE!**
- **Step 2: Particular Solution:**

$$x_p(t) = t(A \cos 3t + B \sin 3t)$$

Substituting into $x'' + 9x$:

$$x_p'' + 9x_p = -6A \sin 3t + 6B \cos 3t = 4 \cos 3t$$

$$-6A = 0 \implies A = 0, \quad 6B = 4 \implies B = \frac{2}{3}$$

$$x_p(t) = \frac{2}{3}t \sin 3t$$

- **Step 3: General Solution:**

$$x(t) = c_1 \cos 3t + c_2 \sin 3t + \frac{2}{3}t \sin 3t$$

- **Step 4: Apply $x(0) = 0, x'(0) = 0$:**

$$x(0) = c_1 = 0$$

$$x'(t) = 3c_2 \cos 3t + \frac{2}{3} \sin 3t + 2t \cos 3t \implies x'(0) = 3c_2 = 0 \implies c_2 = 0$$

- **Step 5: Final Solution:**

$$x(t) = \frac{2}{3}t \sin 3t$$

Notice the amplitude $\frac{2}{3}t$ **grows linearly to infinity** as $t \rightarrow \infty$! This is the physical mechanism that shatters structures and bridges!

3. Common Exam Traps & Professor Leonard Warnings

- **Trap 1: Confusing Natural vs Quasi-Frequency:** $\omega_0 = \sqrt{k/m}$ is undamped frequency. $\omega_d = \sqrt{\omega_0^2 - \gamma^2}$ is damped frequency. $\omega_d < \omega_0$ always!
- **Trap 2: Resonance Amplitude Blow-Up:** In un-damped systems driven at ω_0 , x_p has a factor of t . In damped systems ($c > 0$), friction prevents infinite blow-up!